

Investigating Embedding Structure for Anomaly Detection in Vision Transformers

Summer Research Bursary

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Proposed Timeline

Week	Activity
1	Dataset Analysis
2	Embedding Pipeline
3	Baseline Evaluation
4-5	Embedding Space Analysis
6-7	Statistical Modelling
8-9	Embedding Space Optimisation
10	Evaluation and Write-Up

Dataset Analysis

The first part of this project will focus on understanding the dataset used throughout the study. The MVETC dataset [1] is a widely used benchmark for anomaly detection in industrial inspection tasks and contains multiple object categories, including both normal and defective samples.

The main goals of this stage are:

1. Identify the different object categories present within the dataset (e.g., bottles, cables, capsules, hazelnuts, etc.).
2. Analyse the distribution of normal and anomalous samples within each category and identify any class imbalances.
3. Examine the different defect types present in the dataset (e.g., scratches, cracks, contamination, structural defects) and understand how they are represented visually.
4. Investigate the characteristics of each category, including image resolution, sample count, and defect variability.
5. Identify potential challenges and limitations within the dataset that may influence anomaly detection performance.
6. Choose which categories and defect types to focus on for the subsequent stages of the project based on the analysis.

This stage will provide an understanding of the data and help guide the design of the anomaly detection pipeline used in later stages of the project.

An additional thing that can be looked at later is looking at other anomaly detection datasets to assess how the generalisation of the developed methods and determine whether changes made on MVTec transfer to other domains.

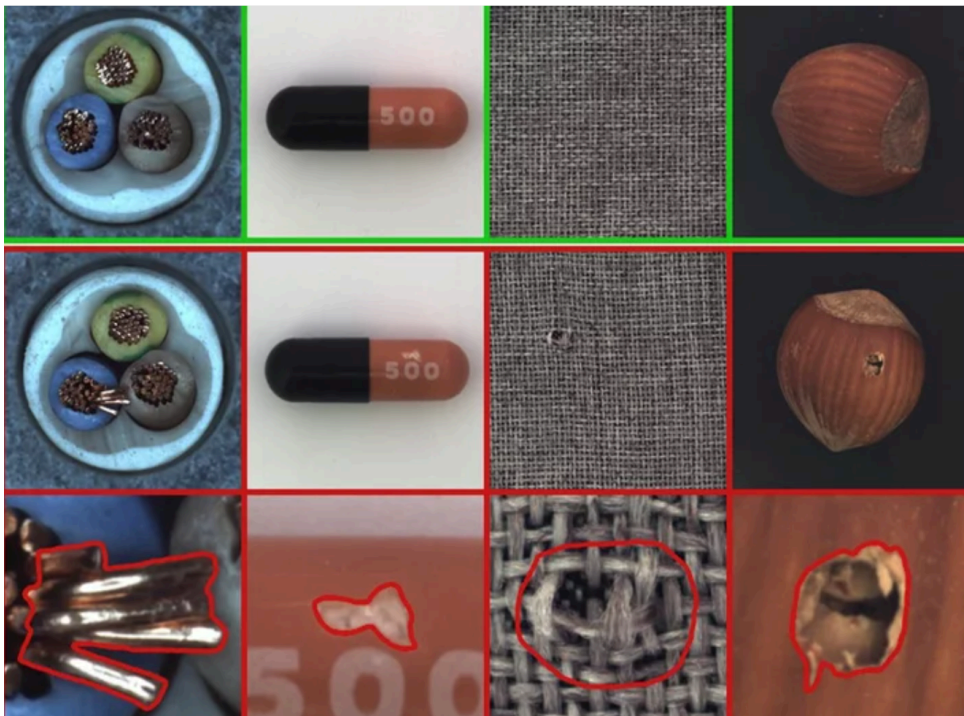


Figure 1: Example images from the MVTec dataset

Embedding Pipeline

This stage of the project will focus on developing a reusable embedding extraction pipeline. The purpose of this pipeline is to avoid repeatedly processing the same images through the Vision Transformer model during experimentation.

Vision Transformer Selection

A pre-trained Vision Transformer will be used as the primary feature-extraction model. The first implementation will focus on DINO [2] family of models due to its strong performance as a general-purpose visual representation model.

Alternative architectures may also be looked at to determine how model choice influences embedding quality and anomaly detection performance.

The dataset will first be organised into a structured metadata table containing information such as image categories, defect types, labels, and train/test splits. This metadata will allow normal and anomalous samples to be queried and compared during experiments.

Each image will then be passed through the model to extract both global and local representations. The global representation will be obtained from the CLS token, while local representations will be obtained from patch-level embeddings. These embeddings will be stored separately so that global, patch-level, and hybrid approaches can be compared later in the project.

The planned storage approach is to use SQLite for metadata and FAISS for embeddings. SQLite will provide a way to query image-level information, while FAISS will support similarity search and nearest-neighbour lookup in embedding space. This structure will make it easier to run experiments repeatedly without recomputing embeddings each time.

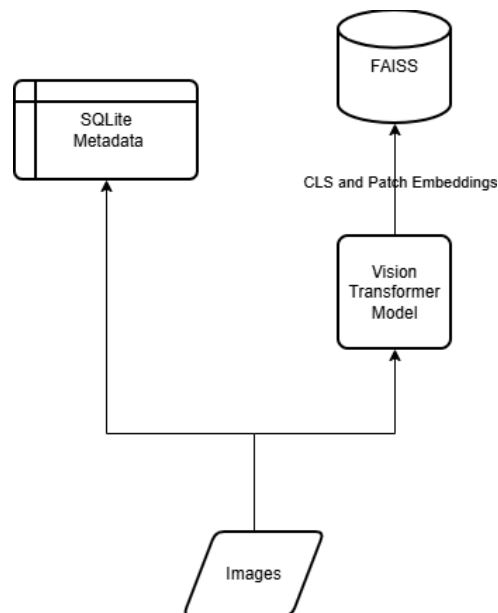


Figure 2: Diagram of the embedding extraction pipeline

Baseline Model

The first stage of experimentation will focus on establishing baseline anomaly detection performance using the CLS token. The objective of this stage is to understand how effectively anomalies can be detected using the learned representations before introducing additional techniques such as fine-tuning, hybrid embeddings, or statistical modelling.

Initially, experiments will be performed using the extracted embeddings, with minimal hyperparameter tuning, focusing on ensuring stable convergence and reproducible results. This will provide a reference point against which later approaches can be compared.

Several distance- and similarity-based methods will be investigated for anomaly scoring, including:

- K-Nearest Neighbours
- Mean Embedding Distance
- Centroid-Based Methods
- Mahalanobis Distance [3]

Hybrid approaches will also be investigated. One potential approach is a hierarchical pipeline in which global CLS embeddings are first used to identify potentially anomalous samples, with patch-level analysis applied only to images that exceed a predefined anomaly threshold. This may improve computational efficiency while retaining the ability to localise anomalies when required.

The performance of each approach will be evaluated using standard anomaly detection metrics, including:

- AUROC & AUPRC

The results obtained during this stage will serve as a baseline for later experiments involving patch-level embeddings, hybrid representations, and embedding-space analysis.

Embedding Space Analysis and Visualisation

This stage of the project will investigate how normal and anomalous samples are distributed within the learned embedding space. The objective is to understand how model representations separate different types of data and to determine whether anomalies form distinct regions or deviations from normal samples.

Analysis will be performed at both the image level, using CLS embeddings, and the local level, using patch embeddings. This will allow investigation of both how entire images relate to one another within the embedding space and how individual patches relate to other patches within the same image and across different images.

Scoring metrics will be used to analyse cluster quality and separation, including:

- Silhouette Score
- Intra-Cluster Distance
- Inter-Cluster Distance

To help interpret these results, visualisation techniques to understand the embedding space include:

- PCA
- UMAP
- t-SNE

These visualisations will be used to investigate cluster compactness, the separation between normal and anomalous samples, and the distribution of different defect types within the embedding space.

Statistical Modelling

This stage of the project will investigate the statistical structure of the learned embedding space and the significance of distances between embeddings.

Covariance-based approaches, such as Mahalanobis distance, will be investigated to model normal regions within the embedding space. The resulting ellipsoidal regions may provide a more interpretable representation of normal behaviour and anomaly detection compared to simple distance-based approaches. Rather than using a fixed-radius threshold, ellipsoidal modelling adapts the boundary of normal behaviour to the underlying data distribution, allowing variation to be represented differently along each principal direction.

The statistical structure of the embedding space will also be analysed using image metadata, such as object category and defect type, to investigate whether different anomalies form distinct regions or patterns within it.

Embedding Space Optimisation

This stage of the project will investigate whether the structure of the embedding space can be improved through fine-tuning and alternative training objectives. The goal is to determine whether more distinct boundaries can be established between normal and anomalous samples, thereby improving anomaly detection performance and interpretability.

Experiments will investigate techniques such as sample mining, alternative objective functions, and embedding-space regularisation. These approaches will be explored to determine whether the learned representations can be made more compact, better separated, and more robust to subtle variations within the data.

Particular attention will be given to the formation of neighbourhoods and clusters within the embedding space, and how normal samples relate to one another compared to anomalous samples. Statistical measures developed in earlier stages of the project will be used to analyse whether fine-tuning leads to improved cluster formation and anomaly separability.

Where anomaly detection performance is limited, experiments will investigate whether fine-tuning can make subtle anomalies more distinguishable within the embedding space. The objective is to increase the separation between normal and anomalous samples while preserving the structure of the normal data.

The project will also explore the use of adaptive or dynamic objective functions. Rather than using a fixed objective throughout optimisation, different training objectives may be emphasised at different stages of training based on the observed structure of the embedding space. This will be investigated as a potential mechanism to improve cluster quality, reduce undesirable embedding-space behaviour, and create more representative statistical boundaries.

Evaluation

The developed methods will be evaluated using a range of anomaly detection, embedding-space, and computational performance metrics.

Anomaly detection performance will be evaluated using:

- AUROC
- AUPRC
- F-beta Scores (depending on the wanted balance between precision and recall)
- Confusion Matrix Metrics (Precision, Recall, Accuracy, and False Positive Rate)

Embedding-space quality will be evaluated using:

- Silhouette Score
- Intra-Cluster Distance
- Inter-Cluster Distance

Where appropriate, computational performance will also be considered, including:

- Embedding extraction time
- Inference time
- Memory requirements

Comparisons will be performed between global, local, and hybrid approaches to determine which representation provides the best balance between anomaly detection performance, interpretability, and computational efficiency.

Extended Work

If time permits towards the end of the bursary, additional work will focus on improving the interpretability of anomaly detection decisions. This will build on the outputs of previous stages of the project and investigate ways to make anomaly detection decisions easier to understand and explain.

Potential areas include:

- Identifying and visualising the patch regions that contribute most strongly to an anomaly decision.
- Highlighting the likely location of defects within an image using patch-level anomaly scores.
- Retrieving similar normal and anomalous samples from the embedding database to provide additional context for anomaly decisions.

A further extension would investigate using a small local language model, such as Mistral, to generate natural-language explanations for anomaly detection. The model could be adapted using LoRA fine-tuning and provided with statistical outputs from the anomaly detection pipeline, including anomaly scores, embedding-space distances, nearest-neighbour results, and patch-level information.

The objective would be to generate human-readable explanations describing why an image was classified as anomalous and which characteristics contributed most strongly to the decision. This could provide an additional layer of interpretability beyond the anomaly scores and visualisations provided.

Bibliography

- [1] M. Software, "The MVTec Anomaly Detection Dataset," [Online]. Available: <https://www.mvtec.com/research-teaching/datasets/mvtec-ad>
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- [3] R. Kamoi and K. Kobayashi, "Why is the Mahalanobis Distance Effective for Anomaly Detection?." [Online]. Available: <https://arxiv.org/abs/2003.00402>